

```

In [11]: # Make dependent variable as a factor (categorical)
data$target = as.factor(data$target)

In [12]: # Splitting the dataset into test and train
print("Train Test Split") # 70/30 Split
dt = sort(sample(nrow(data), nrow(data)*.7))
train<-data[dt,]
val<-data[-dt,]

[1] "Train Test Split"

In [13]: # No. of rows in Train and Val Dataset
nrow(train)
nrow(val)

212
91

```

Fig. 2: Partitioning of the dataset

Loading and Understanding the dataset

```

In [4]: data = read.csv("/content/heart_health.csv")

In [5]: print("Minimum resting blood pressure")
min(data$trestbps)
print("Maximum resting blood pressure")
max(data$trestbps)
print("Summary of Dataset")
summary(data)

[1] "Minimum resting blood pressure"
94
[1] "Maximum resting blood pressure"
200
[1] "Summary of Dataset"

```

age		sex	cp	trestbps
Min. :29.00	Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :94.0
1st Qu.:47.50	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:120.0
Median :55.00	Median :1.0000	Median :1.0000	Median :1.0000	Median :130.0
Mean :54.37	Mean :0.6832	Mean :0.967	Mean :0.967	Mean :131.6
3rd Qu.:61.00	3rd Qu.:1.0000	3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:140.0
Max. :77.00	Max. :1.0000	Max. :3.000	Max. :3.000	Max. :200.0

chol		fbs	restecg	thalach
Min. :126.0	Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :71.0
1st Qu.:211.0	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:133.5
Median :240.0	Median :0.0000	Median :1.0000	Median :1.0000	Median :153.0
Mean :246.3	Mean :0.1485	Mean :0.5281	Mean :0.5281	Mean :149.6
3rd Qu.:274.5	3rd Qu.:0.0000	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:166.0
Max. :564.0	Max. :1.0000	Max. :2.0000	Max. :2.0000	Max. :202.0

exang		oldpeak	slope	ca
Min. :0.0000	Min. :0.00	Min. :0.000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.00	1st Qu.:1.000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.80	Median :1.000	Median :0.0000	Median :0.0000
Mean :0.3267	Mean :1.04	Mean :1.399	Mean :0.7294	Mean :0.7294
3rd Qu.:1.0000	3rd Qu.:1.60	3rd Qu.:2.000	3rd Qu.:1.0000	3rd Qu.:1.0000
Max. :1.0000	Max. :6.20	Max. :2.000	Max. :4.0000	Max. :4.0000

thal		target
Min. :0.000	Min. :0.0000	Min. :0.0000
1st Qu.:2.000	1st Qu.:0.0000	1st Qu.:0.0000
Median :2.000	Median :1.0000	Median :1.0000
Mean :2.314	Mean :0.5446	Mean :0.5446
3rd Qu.:3.000	3rd Qu.:1.0000	3rd Qu.:1.0000

Fig. 3: Outcome of dataset—minimum vs maximum blood pressure

Next, load the dataset:

```

data = read.csv("heart_health.csv")

Make the dependent variable a factor (categorical):

data$target = as.factor(data$target)

Now split the dataset into 'test' and 'train':

print("Train Test Split") # 70/30 Split
dt = sort(sample(nrow(data), nrow(data)*.7))
train<-data[dt,]
val<-data[-dt,] nrow(train) nrow(val)

```

Fig. 2 represents the partitioning, i.e., the splitting of the dataset into train set and test set in a 70 to 30 ratio, respectively.

Fig. 3 helps understand the dataset being considered for the health analysis case study.

To analyse the data in the dataset, type:

```

print("Minimum resting blood pressure")
min(data$trestbps)
print("Maximum resting blood pressure")
max(data$trestbps)
print("Summary of Dataset")
summary(data)

```

To arrive at the range of resting blood pressure, use the following code:

```

print("Range of resting blood pressure")
max(data$trestbps) - min(data$trestbps)
quantile(data$trestbps, c(0.25, 0.5, 0.75))
print("Column name of the Data")
names(data)
print("Attributes of the Data")
str(data)
print("Number of Rows and Columns:")
dim(data)

```

Fig. 4 captures the range of resting blood pressure for the health analysis case study.

To analyse the correlation between resting BP and age, use following code snippet:

```

print("Correlation between the resting blood pressure and the age")
cor(data$trestbps, data$age, method = "pearson")
cor.test(data$trestbps, data$age, method = "pearson")

```

Fig. 5 gives the correlation between resting BP and age.